IAT Test 1

Question paper, answer key
and worked solutions

> **MANUAL REVIEW DRAFT**
> 2027 preparation using the verified 2026 pattern and stored 2026 syllabus.
> The 2027 brochure has not been verified. Not approved for student publication.

60	180	240
single-correct questions	minutes	maximum marks

Subject	Question numbers	Marks
Biology	1-15	60
Chemistry	16-30	60
Mathematics	31-45	60
Physics	46-60	60

Instructions

1. There are 60 single-correct questions: Biology 1–15, Chemistry 16–30, Mathematics 31–45 and Physics 46–60.
2. Each question has exactly one correct option. Correct: +4; incorrect: -1; unanswered: 0. All questions are available to attempt; there is no best-three-subject scoring rule.
3. Use 180 minutes. An on-screen non-programmable scientific calculator is part of the 2026 examination conditions; personal calculators are not allowed under that scheme.
4. Treat stated data and idealisations as exact unless an approximation is explicitly supplied. Log denotes base 10 and ln denotes the natural logarithm.
5. The review PDF contains the question paper first, followed by a separate answer key and worked solutions. Keep the solutions hidden during a timed attempt.

Fixed authored order. Figures and IDs must remain linked during review.
Prepared 29 October 2026 · IAT27_TEST_01 · Original AI-assisted content.

QUESTION PAPER / B

Biology

Questions 1-15 · 15 questions · 60 marks

Q1

+4 / -1

In plant X, the conspicuous green generation produces gametes; the spore-producing generation remains attached to it and nutritionally dependent on it. In plant Y, the conspicuous green generation produces spores and is independent. Which identification is consistent with these observations?

A) X is a fern and Y is a moss

B) X is a moss and Y is a fern

C) Both X and Y must be mosses

D) Both X and Y must be ferns

Q2

+4 / -1

Two young stem sections are examined. Section P has many scattered, conjoint vascular bundles without cambium between xylem and phloem. Section Q has conjoint bundles in a ring, with cambium between xylem and phloem. Under the usual NCERT model of stem anatomy, which prediction is best supported?

A) P should undergo ordinary cambium-driven secondary growth, but Q should not

B) Both sections are roots because xylem and phloem occur in the same bundle

C) Both sections have closed vascular bundles

D) Q can establish a vascular cambium ring for secondary growth; P normally cannot

Q3

+4 / -1

Three equal preparations of an enzyme are assayed at the same final pH and temperature. U is untreated. I contains a reversible inhibitor. H was heated strongly and then cooled.
Approximate initial rates are:
Low substrate: U = 40, I = 20, H = 4 units.
Very high substrate: U = 100, I = 100, H = 10 units.
Rates are rounded to the experimental precision. Which interpretation is consistent with both substrate conditions?

A) I is consistent with competitive inhibition; heating has reduced the amount of functional enzyme in H

B) I must have destroyed half the enzyme molecules irreversibly; H is fully restored by cooling

C) I has doubled the maximum catalytic rate; H has increased substrate affinity

D) Both I and H merely reduce substrate concentration, with no effect on enzyme function

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Q4

+4 / -1

A test-cross of a double heterozygote produces 420 Ab, 420 aB, 80 AB and 80 ab progeny. Assume these frequencies accurately represent gamete frequencies, recombination is the same in male and female meiosis, and there is no viability selection. If two individuals with this same double-heterozygous genotype and phase are crossed, what percentage of their offspring is expected to be aabb?

A) 6.4%

B) 17.64%

C) 0.64%

D) 25%

Q5

+4 / -1

In an idealised kidney experiment, glomerular filtration and the medullary osmotic gradient are kept constant. ADH action on the collecting ducts is then increased. Which immediate combination is expected, assuming the ducts respond normally?

A) Lower collecting-duct water permeability and a larger volume of dilute urine

B) Greater water reabsorption and a smaller volume of more concentrated urine

C) Greater glucose filtration and complete loss of the medullary gradient

D) Unchanged water handling because ADH acts only on glomerular blood flow

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Q6

+4 / -1

An ecosystem's plants fix 2400 kJ m−2 day−1 and use 900 kJ m−2 day−1 in respiration. Herbivores assimilate 180 kJ m−2 day−1 of plant-derived energy and respire 120 kJ m−2 day−1; the remainder becomes new herbivore biomass. What percentage of plant net primary production becomes this new herbivore biomass?

A) 4%

B) 7.5%

C) 12%

D) 2.5%

Q7

+4 / -1

A diploid cell has 12 chromosomes in G1. It completes S phase and reaches G2 without mitosis. If its G1 DNA content is 2C, what are its chromosome number and DNA content in G2? Count chromosomes by centromeres, before sister chromatids separate.

A) 24 chromosomes and 4C DNA

B) 12 chromosomes and 2C DNA

C) 24 chromosomes and 2C DNA

D) 12 chromosomes and 4C DNA

Q8

+4 / -1

In a typical diploid angiosperm, one male gamete successfully fuses with the egg, but the other male gamete fails to fuse with the central cell. Assume normal reduced gametes, two haploid polar nuclei and no autonomous endosperm development. Which immediate outcome follows?

A) A haploid zygote forms together with a triploid primary endosperm nucleus

B) A triploid zygote forms, but a diploid endosperm nucleus does not

C) A diploid zygote forms, but a triploid primary endosperm nucleus does not

D) Neither a zygote nor an embryo sac can exist

Q9

+4 / -1

A sealed thylakoid preparation is placed in darkness with the imposed pH difference shown. Its membrane is otherwise proton-tight, its ATP synthase is functional, and ADP and inorganic phosphate are available outside. No other energy source is supplied. Which short-term observation is expected?

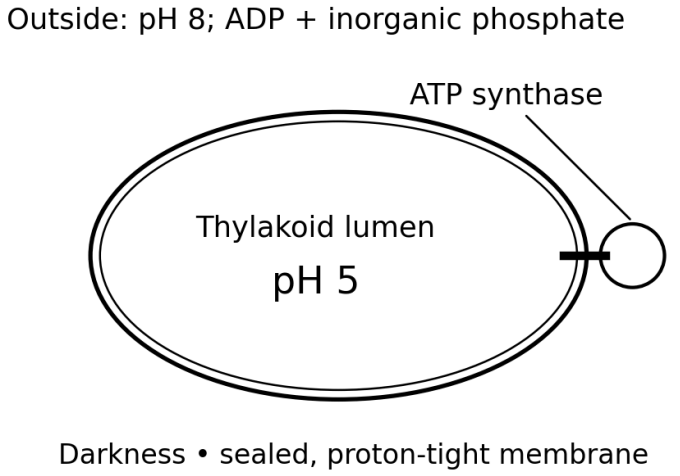


Figure for Q9

A) ATP can be made transiently, but the gradient alone does not drive light-dependent NADPH formation

B) Sustained ATP and NADPH formation continue indefinitely in darkness

C) No ATP can be made because an imposed proton gradient is never an energy source

D) ATP forms only if protons move from the outside into the more acidic lumen

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Q10

+4 / -1

In an immunology comparison, preparation P contains ready-made antitoxin antibodies, whereas preparation Q contains a harmless toxoid that elicits the recipient's own immune response. Which comparison is expected in an immunologically responsive recipient?

A) P and Q both provide memory cells directly

B) P can protect sooner but normally does not establish antigen-specific memory; Q can establish memory after a delay

C) Q protects immediately by supplying antibodies, whereas P requires clonal selection

D) P produces lasting active immunity, whereas Q produces only passive immunity

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Q11

+4 / -1

A diploid locus is PCR-amplified to a 900 bp product. Allele R has one restriction site, giving 600 bp and 300 bp fragments after complete digestion; allele r has no site. The sample lane is shown. Assume complete digestion, equal PCR amplification per allele, equal DNA recovery, no other fragments and fluorescence proportional to DNA mass. What fraction of the sample-lane fluorescence should occur in the 900 bp band?

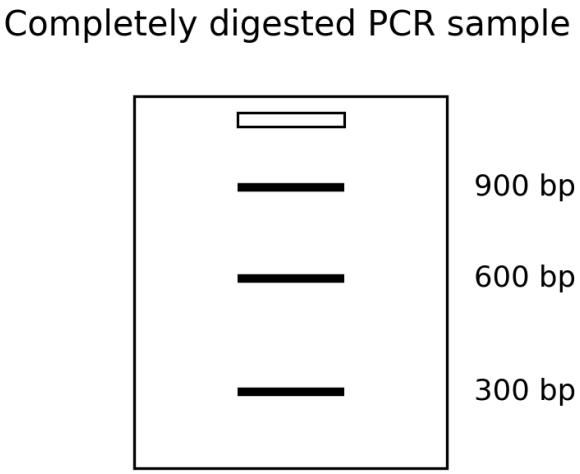


Figure for Q11

A) One-third

B) One-sixth

C) One-half

D) Two-thirds

Q12

+4 / -1

Muscle-cell preparations A and B are viable and have equal basal glucose uptake. Adding insulin increases uptake in A but not in B. A separate binding assay establishes that B's surface insulin receptors cannot bind insulin; downstream uptake machinery is otherwise intact. Which conclusion is best supported?

A) Insulin normally acts by binding directly to nuclear DNA

B) B must have lost every glucose transporter, including those responsible for basal uptake

C) B's failure proves that its source organism cannot produce insulin

D) The defect is at hormone reception, so intact downstream machinery alone does not ensure an insulin response

Q13

+4 / -1

A fully penetrant autosomal-recessive disorder is caused by allele a. An unaffected person has an affected sibling and two unaffected parents; assume the parents are Aa × Aa. This person has a child with an unrelated, unaffected partner from a Hardy–Weinberg population where the frequency of a is 0.20. Assume random mating, no new mutation and independent family origins. What is the probability that the child is affected?

A) 1/12

B) 1/18

C) 1/24

D) 1/6

Q14

+4 / -1

A frog is submerged in well-oxygenated water, so that its lungs are not being ventilated. Its skin remains moist, intact and well supplied with blood. Which statement correctly explains how oxygen uptake can still occur under these conditions?

A) Dissolved oxygen can diffuse across the vascular skin; this does not require the frog to have gills

B) Its skin produces oxygen by a light-dependent process

C) Its pulmonary ventilation continues normally because immersion does not block air entry

D) Oxygen enters only because the frog's heart has four completely separated chambers

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Q15

+4 / -1

An initially neutral allele has frequency 0.50 in each of eight isolated experimental populations. Four populations are very small and four are very large. After many generations, the small populations have allele frequencies 0.00, 0.10, 0.90 and 1.00; the large populations remain near 0.50. Mutation, migration and fitness differences are excluded. Which explanation best fits the replicate outcomes?

A) Directional selection favours the allele equally in every population

B) The allele must have become beneficial in the two populations where it increased

C) Random sampling of alleles has stronger effects in the small populations

D) Hardy–Weinberg proportions require allele frequencies to change identically in all populations

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Chemistry

Questions 16-30 · 15 questions · 60 marks

Q16

+4 / -1

An isolated carbonate ion, $\mathrm{CO_3}^{2-}$, is described using its three equivalent principal Lewis resonance structures. Which statement about its three carbon–oxygen bonds is correct?

A) One bond is permanently double and the other two are permanently single

B) The ion rapidly alternates between three distinguishable bond arrangements

C) All three bonds have bond order 2

D) All three bonds are equivalent, with average bond order 4/3 in this resonance model

Q17

+4 / -1

Compare octahedral $[\mathrm{Co(NH_3)_6}]^{3+}$ and $[\mathrm{Co(CN)_6}]^{3-}$. Assume the observed absorption in each comparison is the transition associated with its octahedral splitting Δ_o. Cyanide produces the larger splitting. Which statement follows?

A) The cyanide complex absorbs at shorter wavelength, while cobalt is +3 in both complexes

B) The cyanide complex absorbs at longer wavelength, while cobalt is +3 in both complexes

C) The two complexes must absorb at the same wavelength because both have six ligands

D) The cobalt oxidation states are +3 and −3, respectively, so ligand strength is irrelevant

Q18

+4 / -1

For a reaction, $\Delta H = +24.0\ \mathrm{kJ\,mol^{-1}}$ and $\Delta S = +80.0\ \mathrm{J\,K^{-1}mol^{-1}}$, independent of temperature over the range considered. Which pair gives ΔG at 200 K and 400 K, respectively?

A) −8.0 and +8.0 kJ mol−1

B) +40.0 and +56.0 kJ mol−1

C) +8.0 and −8.0 kJ mol−1

D) +23.984 and +23.968 kJ mol−1

Q19

+4 / -1

A mixture contains only propanal and propanone. Separate complete tests on identical portions of the mixture give 2.16 g of silver with Tollens reagent and 3.94 g of iodoform in the iodoform test. Propanal gives no iodoform under these conditions. Take $M(\mathrm{Ag}) = 108\ \mathrm{g\,mol^{-1}}$ and $M(\mathrm{CHI_3}) = 394\ \mathrm{g\,mol^{-1}}$. What is the mole fraction of propanone in the mixture?

A) 1/3

B) 1/2

C) 2/3

D) 1/4

Q20

+4 / -1

A zinc–copper galvanic cell has $E^\circ_{\mathrm{cell}} = 1.10\ \mathrm{V}$ and $[\mathrm{Zn^{2+}}] = 1.0\ \mathrm{M}$. In the copper half-cell, solid $\mathrm{Cu(OH)_2}$ is present and a buffer maintains $[\mathrm{OH^-}] = 10^{-4}\ \mathrm{M}$. Given $K_{sp}(\mathrm{Cu(OH)_2}) = 10^{-20}$, estimate the cell emf. Treat activities as concentrations, neglect junction potentials, and take $2.303RT/F = 0.060\ \mathrm{V}$.

A) 1.46 V

B) 1.10 V

C) 0.38 V

D) 0.74 V

Q21

+4 / -1

A mixture of 0.010 mol methylamine and 0.020 mol aniline is treated with sufficient nitrous acid at 0–5 °C. Assume quantitative reactions: the primary aliphatic amine evolves nitrogen, while the aromatic diazonium salt remains stable in the cold. The remaining mixture is then warmed so that all of the diazonium salt decomposes, evolving nitrogen. Which pair gives the moles of nitrogen evolved in the cold stage and the additional moles on warming?

A) 0.030 mol; 0 mol

B) 0.020 mol; 0.010 mol

C) 0.010 mol; 0.020 mol

D) 0 mol; 0.030 mol

Q22

+4 / -1

Treat $\mathrm{Fe_3O_4}$ as containing only $\mathrm{Fe^{2+}}$, $\mathrm{Fe^{3+}}$ and $\mathrm{O^{2-}}$. What fraction of its iron ions must be in the +3 oxidation state?

A) 2/3

B) 1/3

C) 1/2

D) 1

Q23

+4 / -1

A first-order reaction is 25% complete after 10.0 min. How much additional time is required for the reaction to become 75% complete? Take $\ln(4/3) = 0.288$ and $\ln 3 = 1.099$.

A) 20.0 min

B) 38.2 min

C) 48.2 min

D) 30.0 min

Q24

+4 / -1

Racemic $\mathrm{CH_3COCH(CH_3)CH{=}CH_2}$ is treated with $\mathrm{NaBH_4}$, reducing only the ketone to an alcohol. Both faces of the carbonyl can react. After work-up, all the alcohol is acylated with ethanoyl chloride in pyridine. The terminal C=C bond survives, and no stereocentre subsequently changes configuration. What are the molecular formula and maximum number of distinct configurational stereoisomers of the final ester? Count all products accessible from both starting enantiomers; ignore conformers.

A) $\mathrm{C_8H_{12}O_2}$; 4

B) $\mathrm{C_8H_{14}O_2}$; 2

C) $\mathrm{C_8H_{14}O_2}$; 4

D) $\mathrm{C_8H_{14}O_2}$; 8

Q25

+4 / -1

A sample containing only the two enantiomers of butan-2-ol has zero optical rotation in an achiral solvent. It is oxidised completely to butan-2-one, which is then reduced with an achiral reagent in an achiral environment. No resolution is performed. Which statement is correct?

A) The starting sample is optically inactive because every butan-2-ol molecule is achiral

B) Oxidation must preserve the original stereocentre in butan-2-one

C) Reduction must produce only one enantiomer because the starting alcohol was optically inactive

D) The starting sample is racemic, and reduction of the planar ketone is expected to give a racemic alcohol mixture

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Q26

+4 / -1

Two solutions at temperature T are separated by an ideal membrane permeable only to water. The left side contains 0.10 M sucrose. The right side contains 0.060 M $\mathrm{CaCl_2}$, of which 75% is dissociated; assume ideal behaviour otherwise. Which initial water-flow direction and pressure difference would prevent that flow? Use R in L atm mol−1 K−1.

A) Water flows right to left; apply an extra 0.040RT atm on the left

B) Water flows left to right; apply an extra 0.050RT atm on the right

C) Water flows left to right; apply an extra 0.080RT atm on the right

D) No initial flow occurs because both solutes have the same mass concentration

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Q27

+4 / -1

A 1.90 g mixture contains only anhydrous $\mathrm{Na_2CO_3}$ and $\mathrm{NaHCO_3}$. It is heated until all bicarbonate decomposes, leaving 1.59 g of anhydrous solid. What volume of 1.00 M HCl is required to neutralise this entire residue completely to $\mathrm{CO_2}$, water and sodium chloride? Take $M(\mathrm{Na_2CO_3}) = 106\ \mathrm{g\,mol^{-1}}$.

A) 30.0 mL

B) 15.0 mL

C) 60.0 mL

D) 35.8 mL

Q28

+4 / -1

Consider the ions $\mathrm{S^{2-}}$, $\mathrm{Cl^{-}}$, $\mathrm{K^{+}}$ and $\mathrm{Ca^{2+}}$. Their atomic numbers are 16, 17, 19 and 20, respectively. Which order of ionic radii is expected for these isoelectronic ions?

A) Ca2+ > K+ > Cl− > S2−

B) K+ > Ca2+ > S2− > Cl−

C) Cl− > S2− > Ca2+ > K+

D) S2− > Cl− > K+ > Ca2+

Q29

+4 / -1

One mole of sucrose is hydrolysed completely to its two monosaccharides. Under ideal quantitative Tollens-test conditions, both resulting monosaccharides act as reducing sugars, each reducing two moles of $\mathrm{Ag^{+}}$. What is the maximum amount of metallic silver formed from the hydrolysate?

A) 0 mol

B) 2 mol

C) 4 mol

D) 1 mol

Q30

+4 / -1

100 mL of 0.40 M ammonia is mixed with 100 mL of 0.10 M HCl. Then 0.020 mol more HCl is added in a small volume. Assuming quantitative neutralisation followed by weak-base equilibrium, what is the final pH? Take $K_b(\mathrm{NH_3}) = 10^{-5}$, $pK_w = 14.00$ and $\log 3 = 0.477$. Activity effects may be neglected.

A) 8.52

B) 9.48

C) 5.00

D) 7.00

QUESTION PAPER / M

Mathematics

Questions 31-45 · 15 questions · 60 marks

Q31

+4 / -1

For $A = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$, what is the sum of the diagonal entries of A^{-1}?

A) 4

B) 1/2

C) 2

D) -2

Q32

+4 / -1

Six distinct folders A, B, C, D, E and F are arranged in a row. A must be to the left of B, with exactly one folder between A and B. How many arrangements satisfy these conditions?

A) 48

B) 96

C) 192

D) 120

Q33

+4 / -1

A function $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(x) + 2f(1 - x) = x^2$ for every real x. What is the minimum value of f on $[-1, 2]$?

A) 0

B) 2/3

C) $-1/3$

D) $-2/3$

Q34

+4 / -1

Two complex numbers z_1 and z_2 are exactly the solutions of $|z - 1| = |z + 1|$ and $|z - 2i| = \sqrt{5}$. What is $\mathrm{Im}(z_1)\,\mathrm{Im}(z_2)$?

A) -1

B) 1

C) -5

D) 4

Q35

+4 / -1

Evaluate $I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x}\, dx$.

A) π/2

B) π²/2

C) π²/4

D) π²/8

Q36

+4 / -1

Point P has coordinates (1,2,3). Its reflection in the plane $x + y + z = 3$ is P′. What is the distance of P′ from the origin?

A) √14

B) √5

C) 2√3

D) √2

Q37

+4 / -1

Real numbers a and b are chosen so that $f(x) = |x^2 - 1| + a|x - 1| + b|x + 1|$ is differentiable at every real x. What is a+b?

A) 4

B) −4

C) −2

D) 0

Q38

+4 / -1

Two groups each contain four observations. Group A has mean 2 and population variance 1; group B has mean 6 and population variance 9. Here each variance uses the number of observations, not one less, as its denominator. What is the population variance of all eight observations together?

A) 9

B) 5

C) 13

D) 18

Q39

+4 / -1

Find $\lim_{x \to 0} \dfrac{\sqrt{1+3x} - \sqrt{1-x}}{x}$.

A) 4

B) 1

C) 0

D) 2

Q40

+4 / -1

Six distinct keys are assigned randomly, one each, to six numbered locks. Exactly one key is correct for each lock, and every one of the $6!$ assignments is equally likely. What is the probability that exactly two locks receive their correct keys?

A) 1/8

B) 1/48

C) 3/16

D) 1/2

Q41

+4 / -1

Using principal values of inverse tangent, evaluate $\tan^{-1} 2 + \tan^{-1} 3$.

A) 3π/4

B) −π/4

C) π/4

D) tan−1 5

Q42

+4 / -1

For $x > 0$, a function satisfies $\dfrac{dy}{dx} = \dfrac{y}{x(1+x)}$ and y(1)=2. What is y(3)?

A) 3/2

B) 3

C) 6

D) 4

Q43

+4 / -1

For which real values of a does $x^2 - 2ax + a + 2 = 0$ have two real roots r1 and r2 satisfying $r_1 < 1 < r_2$?

A) a ≥ 3

B) a < −1 or a > 2

C) a > 3

D) a < 3

Q44

+4 / -1

For $0 \leq a \leq 1$, define $F(a) = \int_0^1 |x^2 - a|\, dx$. What is the minimum possible value of F(a)?

A) 1/3

B) 1/6

C) 1/2

D) 1/4

Q45

+4 / -1

Unit vectors $\mathbf{a}$ and $\mathbf{b}$ make an angle of 60°. A parallelogram has adjacent side vectors $\mathbf{a} + 2\mathbf{b}$ and $\mathbf{a} - \mathbf{b}$. What is its area?

A) √3/2

B) 3√3/2

C) 3

D) 3√3/4

Physics

Questions 46-60 · 15 questions · 60 marks

Q46

+4 / -1

A particle moves along a straight line with the velocity-time graph shown. What are its average speed and average velocity, respectively, over the full 6 s interval?

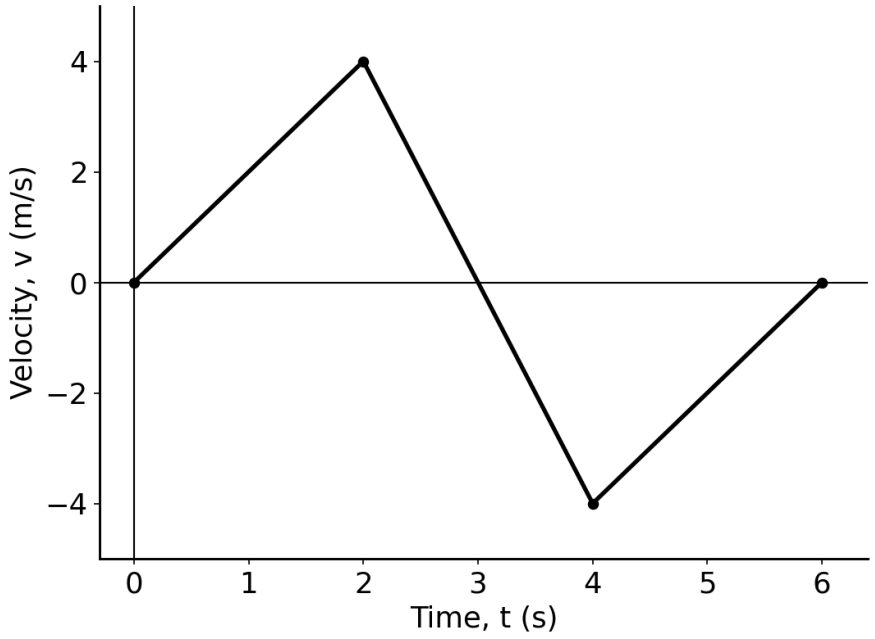


Figure for Q46

A) $2\,\mathrm{m\,s^{-1}},\ 0$

B) $0,\ 2\,\mathrm{m\,s^{-1}}$

C) $4/3\,\mathrm{m\,s^{-1}},\ 0$

D) $2\,\mathrm{m\,s^{-1}},\ 2\,\mathrm{m\,s^{-1}}$

Q47

+4 / -1

An empirical pressure model has the form $P = a\rho v^2 + bh$, where density is ρ, speed is v, height is h, and a is dimensionless. What are the dimensions of b?

A) $ML^{-1}T^{-2}$

B) LT^{-2}

C) $ML^{-2}T^{-2}$

D) ML^{-3}

Q48

+4 / -1

An incompressible, nonviscous fluid of density ρ flows steadily through a horizontal pipe. A wide section has radius r and speed v; a narrow section has radius r/2. What is the pressure in the wide section minus that in the narrow section?

A) $\frac{3}{2}\rho v^2$

B) $\frac{15}{2}\rho v^2$

C) $8\rho v^2$

D) $4\rho v^2$

Q49

+4 / -1

A horizontal disc has constant moment of inertia $2mR^2$ about its frictionless vertical axle. A point bead of mass m initially at radius R co-rotates with it at angular speed ω. A powered actuator attached to the disc moves the bead slowly along a radial groove to R/2. Initially and finally the bead has no radial velocity and co-rotates with the disc. The actuator has negligible inertia; there is no external torque about the axle and no mechanical dissipation. What are the final angular speed and the work supplied by the actuator to the disc-bead mechanical motion?

A) $3\omega/4,\ -mR^2\omega^2/2$

B) $4\omega/3,\ 0$

C) $\omega,\ mR^2\omega^2/2$

D) $4\omega/3,\ mR^2\omega^2/2$

Q50

+4 / -1

A cyclic heat engine absorbs 800 J from a reservoir at 500 K, delivers 300 J of work and rejects the remaining heat to a reservoir at 300 K. Which statement is correct?

A) These figures are consistent with an irreversible engine.

B) These figures describe a reversible engine.

C) These figures necessarily violate the first law of thermodynamics.

D) Its efficiency exceeds the Carnot limit for these reservoirs.

Q51

+4 / -1

A mass m hangs at rest from an ideal massless vertical spring, stretching it by 10 cm. The mass is raised by 5 cm from this equilibrium position and released from rest. With no damping, what is the maximum tensile force in the spring during the resulting motion?

A) mg/2

B) 3mg/2

C) 2mg

D) mg

Q52

+4 / -1

An unknown tuning fork produces 4 beats per second with a 512 Hz reference fork. A small piece of wax is then added only to the reference fork, lowering its frequency to 510 Hz. The beat frequency becomes 2 Hz. What is the frequency of the unchanged unknown fork?

A) 506 Hz

B) 510 Hz

C) 508 Hz

D) 516 Hz

Q53

+4 / -1

A positive charge q moves along the x axis under only the electric force. The field component is shown; $E_0 > 0$ and $a > 0$. At x = 0 the charge moves towards increasing x with kinetic energy $5qE_0a/4$. At what coordinate does it first come to rest?

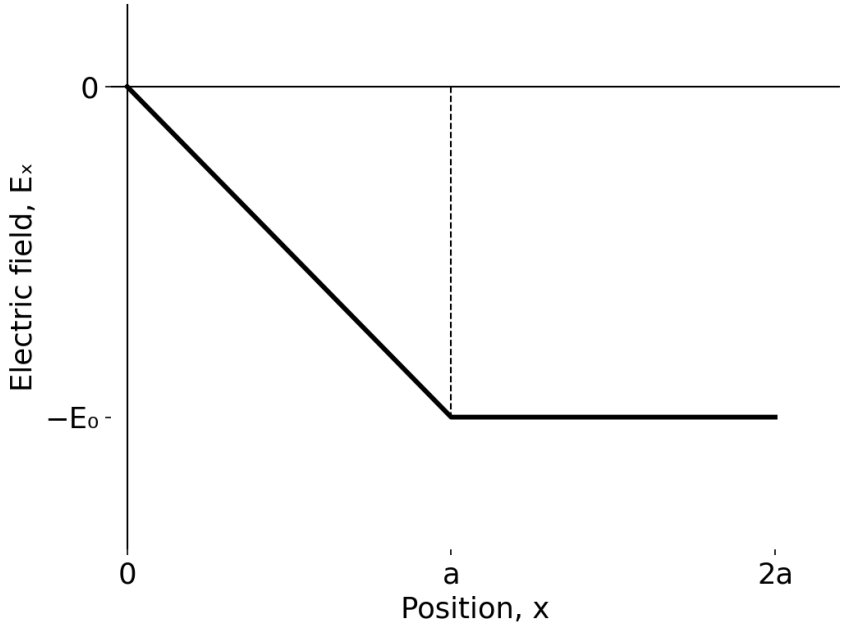


Figure for Q53

A) 5a/4

B) 3a/2

C) 2a

D) 7a/4

Q54

+4 / -1

In the circuit shown, the top rail is 12 V above the bottom rail. The source and ammeter are ideal; all other wires have negligible resistance. What current flows through the ammeter?

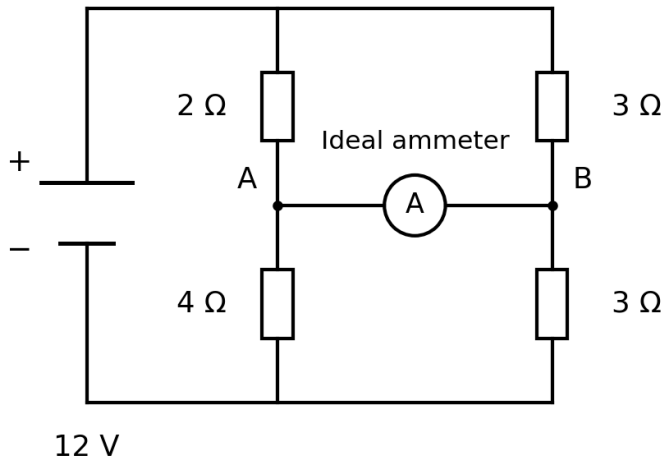


Figure for Q54

A) 2 A, from A to B

B) $12/17$ A, from A to B

C) $12/17$ A, from B to A

D) Zero

Q55

+4 / -1

A nucleus initially has mass number 238 and atomic number 92. It undergoes one alpha decay followed by two beta-minus decays, with no other nuclear transformation. What are its final mass number and atomic number, respectively?

A) 234, 92

B) 234, 90

C) 238, 94

D) 236, 92

Q56

+4 / -1

A conducting bar of mass m and length l slides down two parallel conducting rails inclined at angle θ to the horizontal. A uniform magnetic field of magnitude B is perpendicular to the rail plane. The closed circuit has constant total resistance R. Neglect friction, inductance and all other drag. What is the terminal speed of the bar?

A) $mgR\cos\theta/(B^2l^2)$

B) $mgR\sin\theta/(2B^2l^2)$

C) $2mgR\sin\theta/(B^2l^2)$

D) $mgR\sin\theta/(B^2l^2)$

Q57

+4 / -1

A coin lies at the bottom of the two transparent, immiscible liquid layers shown. It is viewed from air close to the normal. Neglect container thickness. What is its apparent depth below the top liquid surface, and would reversing the order of the layers (keeping each layer's depth) change this apparent depth?

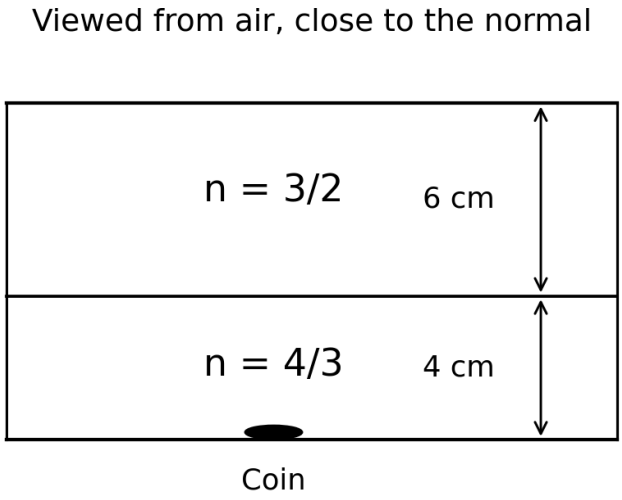


Figure for Q57

A) 10 cm; unchanged

B) 7 cm; changes

C) 7 cm; unchanged

D) $43/3$ cm; unchanged

Q58

+4 / -1

A series circuit contains an ideal resistor, inductor and capacitor driven by a sinusoidal source. The rms current is 2 A, and the rms voltage magnitudes across R, L and C are 8 V, 10 V and 4 V, respectively. What are the source rms voltage and average power supplied?

A) 10 V, 16 W

B) 22 V, 44 W

C) 10 V, 20 W

D) 14 V, 28 W

Q59

+4 / -1

An ideal diode and a resistor R form a half-wave rectifier driven by $v(t) = V_0 \sin \omega t$. There is no smoothing capacitor. What is the ratio of average power dissipated in R to the average power that the same sinusoidal source would dissipate in R if R were connected directly to it? Neglect source resistance and diode voltage drop.

A) $1/\sqrt{2}$

B) 1/2

C) 1/4

D) 1

Q60

+4 / -1

An ideal interferometer uses light of vacuum wavelengths 600 nm and 1000 nm. For each colour the two interfering beams are coherent and have equal amplitudes. At a reference point the geometric optical paths are initially equal, but one beam has an additional phase of π for each colour, so both colours are dark there. A plane glass slab is inserted normally in one path, replacing an equal thickness of air without otherwise changing the geometry. Its refractive index is 1.50 for both colours. What is the smallest positive slab thickness that makes the reference point an interference maximum for both colours?

A) 0.6 µm

B) 1.2 µm

C) 1.8 µm

D) 3.0 µm

Answer key

One correct option per question. Question and option order are fixed.

Biology

Q	Answer
1	**B**
2	**D**
3	**A**
4	**C**
5	**B**
6	**A**
7	**D**
8	**C**
9	**A**
10	**B**
11	**C**
12	**D**
13	**B**
14	**A**
15	**C**

Chemistry

Q	Answer
16	**D**
17	**A**
18	**C**
19	**B**
20	**D**
21	**C**
22	**A**
23	**B**
24	**C**
25	**D**
26	**B**
27	**A**
28	**D**
29	**C**
30	**A**

Mathematics

Q	Answer
31	**C**
32	**B**
33	**D**
34	**A**
35	**C**
36	**D**
37	**B**
38	**A**
39	**D**
40	**C**
41	**A**
42	**B**
43	**C**
44	**D**
45	**B**

Physics

Q	Answer
46	**A**
47	**C**
48	**B**
49	**D**
50	**A**
51	**B**
52	**C**
53	**D**
54	**B**
55	**A**
56	**D**
57	**C**
58	**A**
59	**B**
60	**D**

Reviewer checklist

Accuracy and single correct answer · Syllabus fit · Distractor quality
Diagram labels · Solution clarity · Difficulty and timed attempt

Reviewer: _____ Date: _____

Decision / notes: _____

Editorial difficulty mix: 9 easy / 39 medium / 12 hard. These labels require human review and are not official item statistics.

Biology

Method, reasoning and common errors

Q1

Biology · easy

Diversity In the Living World / Plant kingdom: bryophytes and pteridophytes; dominant generations

Answer: (B) X is a moss and Y is a fern.

Why: A moss has a dominant gametophyte and a dependent, attached sporophyte. In a fern the conspicuous, independent leafy plant is the sporophyte. Gamete versus spore production identifies the generations before the plants are assigned.

Why not the others:

(A) Reverses the characteristic dominance and dependence.

(C) Does not explain Y's dominant independent sporophyte.

(D) Does not explain X's dependent sporophyte.

Key point: Identify the generation by what it produces, then use its independence to distinguish the plant groups.

Q2

Biology · medium

Structural Organisation In Plants And Animals / Anatomy of flowering plants: monocot and dicot stems; open and closed bundles

Answer: (D) Q can establish a vascular cambium ring; P normally cannot.

Why: Q has open bundles, and its intrafascicular cambium can join interfascicular cambium during typical dicot-stem secondary growth. P describes the usual monocot stem with scattered closed bundles. The prediction concerns ordinary secondary growth, not exceptional monocot growth mechanisms.

Why not the others:

(A) Reverses the cambium evidence.

(B) Typical roots have radial vascular arrangement, not the described conjoint stem bundles.

(C) Q explicitly contains cambium and is therefore open.

Key point: Bundle position and cambium presence are separate observations; use both.

Q3

Biology · medium

Cell: Structure And Functions / Biomolecules: enzyme inhibition and effects of temperature

Answer: (A) Competitive inhibition is consistent with I, while H has lost functional enzyme.

Why: At low substrate, I slows the reaction, but excess substrate restores approximately the untreated maximum rate. That is the competitive-inhibition pattern. H remains near one-tenth of U at both conditions, consistent with persistent heat denaturation reducing active enzyme. These results support an interpretation rather than uniquely proving a molecular inhibitor mechanism.

Why not the others:

(B) Irreversible loss would not be rescued by excess substrate; H is not restored.

(C) I never exceeds U's maximum, and H's low maximum does not show improved affinity.

(D) Substrate excess does not rescue H.

Key point: Distinguish substrate rescue from recovery of the enzyme's maximum activity.

Q4

Biology · hard

Genetics And Evolution / Principles of inheritance and variation: linkage, recombination and gamete probabilities

Answer: (C) 0.64%.

Given: 1000 test-cross offspring; ab occurs 80 times.

Formula/Principle: A test-cross reads out the heterozygote's gametes. The aabb offspring must receive ab from each parent.

Step 1: The abundant classes Ab and aB are parental; the recombination fraction is (80 + 80)/1000 = 0.16.

Step 2: Each parent makes ab gametes with probability 80/1000 = 0.08.

Step 3: Independent gamete union gives P(aabb) = 0.08 × 0.08 = 0.0064 = 0.64%.

Common mistake: (A) is ten times too large; (B) squares a parental-gamete frequency; (D) treats the linked two-locus outcome as a single-gene recessive event.

Q5

Biology · medium

Human Physiology / Excretory products and their elimination: ADH and water reabsorption

Answer: (B) Greater water reabsorption and less, more concentrated urine.

Why: ADH increases water permeability in the relevant distal nephron/collecting-duct regions. Water can then move into the hyperosmotic medulla, whose gradient is explicitly maintained. The filtrate loses water without a corresponding proportional removal of every solute.

Why not the others:

(A) Describes the opposite permeability change.

(C) Filtration is controlled and ADH does not imply increased glucose filtration.

(D) Ignores the direct tubular action of ADH.

Key point: Separate a hormone's effect on tubular permeability from changes in filtration.

Q6

Biology · medium

Ecology / Ecosystem: gross/net productivity and trophic energy budgets

Answer: (A) 4%.

Given: Plant gross production 2400; plant respiration 900; herbivore assimilation 180 and respiration 120, all in kJ m−2 day−1.

Formula/Principle: NPP = GPP − producer respiration; net secondary production = assimilation − consumer respiration.

Step 1: Plant NPP = 2400 − 900 = 1500 kJ m−2 day−1.

Step 2: New herbivore biomass stores 180 − 120 = 60 kJ m−2 day−1.

Step 3: The requested percentage is (60/1500) × 100 = 4%.

Common mistake: (B) uses assimilation/GPP; (C) uses assimilation/NPP before consumer respiration; (D) uses consumer production/GPP instead of NPP.

Q7

Biology · easy

Cell: Structure And Functions / Cell cycle and cell division: chromosome number and DNA replication

Answer: (D) 12 chromosomes and 4C DNA.

Why: S phase duplicates each DNA molecule to create sister chromatids. It doubles DNA content but does not yet separate centromeres into daughter chromosomes. The cell remains diploid in G2.

Why not the others:

(A) Counts chromatids as separate chromosomes before centromere separation.

(B) Omits DNA replication.

(C) Increases chromosome count while failing to double DNA.

Key point: DNA quantity and chromosome number need not change together.

Q8

Biology · medium

Reproduction / Sexual reproduction in flowering plants: double fertilisation and triple fusion

Answer: (C) A diploid zygote forms without formation of a triploid primary endosperm nucleus.

Why: Egg-male gamete fusion is n + n = 2n and has succeeded. The separate fusion involving the second male gamete and the two polar nuclei would produce 3n, but that event has failed. The central cell may contain its own diploid secondary nucleus; that is not a fertilised triploid endosperm nucleus.

Why not the others:

(A) Assigns the wrong ploidy to the zygote and invents successful triple fusion.

(B) Interchanges embryo and endosperm ploidies.

(D) Ignores both the pre-existing embryo sac and the stated successful syngamy.

Key point: Double fertilisation consists of two distinct fusion events.

Q9

Biology · medium

Plant Physiology / Photosynthesis in higher plants: chemiosmosis and ATP synthesis

Answer: (A) Transient ATP synthesis is possible without light-dependent NADPH formation.

Why: The lumen has higher proton concentration than the outside. Proton movement down this gradient through ATP synthase can power phosphorylation outside the vesicle. The finite imposed gradient dissipates, so ATP production is transient. A gradient by itself does not provide the light-driven electron-transfer process that reduces NADP+.

Why not the others:

(B) Invents an unlimited energy supply and sustained photochemistry in darkness.

(C) Contradicts chemiosmotic coupling.

(D) Reverses the downhill direction of proton movement.

Key point: The proton gradient is an intermediate energy store; light normally establishes it, but an imposed gradient can temporarily substitute.

Q10

Biology · easy

Biology In Human Welfare / Human health and disease: passive and active immunity

Answer: (B) P acts sooner without normally establishing memory; Q can establish memory after a delay.

Why: Ready-made antibodies provide passive protection while they persist. A toxoid acts as an antigen and stimulates the recipient's adaptive response, including memory formation. This distinction concerns the mechanism of the preparations, not a treatment recommendation.

Why not the others:

(A) Neither preparation supplies ready-made memory cells.

(C) Reverses the source of the antibodies.

(D) Reverses active and passive immunity.

Key point: Receiving antibodies and making one's own immune response are different processes.

Q11

Biology · hard

Biotechnology / Biotechnology: principles and processes; restriction digestion, PCR and electrophoresis

Answer: (C) One-half.

Given: The sample contains bands at 900, 600 and 300 bp; complete digestion and equal amplification are stipulated.

Formula/Principle: R contributes two cut fragments; r contributes one uncut fragment. Fluorescence is proportional to base-pair mass, not simply to the number of bands.

Step 1: All three bands imply an Rr sample under the stated assumptions.

Step 2: For one equally amplified copy of each allele, band masses are proportional to 900, 600 and 300.

Step 3: The 900 bp fraction is 900/(900 + 600 + 300) = 1/2.

Common mistake: (A) gives every band equal mass; (B) is the 300 bp mass fraction; (D) does not conserve the equal 900 bp contribution from the two alleles.

Q12

Biology · medium

Human Physiology / Chemical coordination and integration: insulin action and feedback

Answer: (D) The defect is at hormone reception.

Why: Insulin is a peptide hormone whose action begins at membrane receptors. The binding assay locates the stated defect upstream of the intracellular response. Basal uptake shows that failure of the stimulated response is not equivalent to total absence of glucose uptake.

Why not the others:

(A) Assigns insulin the direct intracellular-receptor mechanism characteristic of a different hormone class.

(B) Contradicts the measured basal uptake.

(C) Confuses reception of an experimentally supplied hormone with pancreatic secretion.

Key point: A failed response can result from a receptor defect even when hormone is available and downstream machinery remains functional.

Q13

Biology · hard

Genetics And Evolution / Principles of inheritance and variation; Hardy–Weinberg principle and conditional carrier risk

Answer: (B) 1/18.

Given: The first parent is an unaffected child of Aa × Aa; the partner is unaffected and q = 0.20.

Formula/Principle: Condition genotype probabilities on being unaffected before multiplying transmission risks.

Step 1: The first person's carrier probability is (1/2)/(3/4) = 2/3.

Step 2: In the population, p = 0.80, so carrier frequency 2pq = 0.32 and unaffected frequency $1 - q^2 = 0.96$. The partner's conditional carrier probability is 0.32/0.96 = 1/3.

Step 3: Two carriers have an affected child with probability 1/4. Overall risk = (2/3)(1/3)(1/4) = 1/18.

Common mistake: (A) drops the first conditioning factor; (C) uses a 1/2 family carrier risk; (D) omits the partner's carrier uncertainty.

Q14

Biology · medium

Structural Organisation In Plants And Animals / Structural organisation in animals: amphibian-frog; respiration

Answer: (A) Cutaneous oxygen diffusion can continue.

Why: Frog skin is a moist, vascular respiratory surface, allowing dissolved oxygen to diffuse into the circulation. The question explicitly removes lung ventilation without removing skin exchange. Whether this supplies all metabolic demand in every situation is a separate issue.

Why not the others:

(B) Frog skin is not photosynthetic.

(C) Contradicts the stipulated absence of lung ventilation.

(D) Adult frogs have a three-chambered heart; the claimed condition is unnecessary for skin respiration.

Key point: Loss of one respiratory route does not automatically eliminate exchange across another functional surface.

Q15

Biology · medium

Genetics And Evolution / Evolution: genetic drift, selection and population size

Answer: (C) Genetic drift is stronger in the small populations.

Why: With selection, migration and mutation excluded, finite sampling of parental alleles can change allele frequency. Opposite directions among replicate small populations, including loss and fixation, fit drift. Large populations show smaller sampling fluctuations.

Why not the others:

(A) Predicts a common selected direction and contradicts the controls.

(B) Infers a fitness advantage that the design excludes.

(D) Confuses genotype proportions under an ideal model with identical realised allele-frequency trajectories.

Key point: Replicate variability and population size can distinguish drift from a consistent selective advantage.

Chemistry

Method, reasoning and common errors

Q16

Chemistry · easy

Chemical Bonding and Molecular Structure / Kössel-Lewis structures, resonance and bond parameters

Answer: (D) Equivalent bonds with average bond order $4/3$.

Why: Each principal contributor contains one C=O bond and two C–O bonds. The resonance hybrid delocalises the bonding across equivalent oxygen positions; it is not a molecule switching between drawings. The model's average is $(2+1+1)/3 = 4/3$.

Why not the others:

(A) Incorrectly localises the double bond.

(B) Treats resonance contributors as separately interconverting molecules.

(C) Gives the model six bond-order units instead of four.

Key point: Resonance drawings are alternative representations of one delocalised structure.

Q17

Chemistry · medium

Coordination compounds / Bonding in coordination compounds: ligand-field splitting and absorption

Answer: (A) Shorter wavelength for the cyanide complex; Co is +3 in both.

Why: In the stated transition model, $\Delta_o = h\nu = hc/\lambda$. A larger splitting therefore requires higher frequency and shorter wavelength. Ammonia is neutral. In the cyanide complex, $x + 6(-1) = -3$, giving $x = +3$.

Why not the others:

(B) Reverses the energy–wavelength relationship.

(C) Coordination number alone does not fix splitting.

(D) Mistakes the complex charge for the oxidation state of the central metal.

Key point: Determine metal oxidation state by charge balance, and spectral energy by ligand-field splitting.

Q18

Chemistry · medium

Thermodynamics / Spontaneity and Gibbs energy change

Answer: (C) $+8.0$ and $-8.0\ \mathrm{kJ\,mol^{-1}}$.

Given: $\Delta H = 24.0\ \mathrm{kJ\,mol^{-1}}$; $\Delta S = 0.0800\ \mathrm{kJ\,K^{-1}mol^{-1}}$.

Formula/Principle: $\Delta G = \Delta H - T\Delta S$.

Step 1: At 200 K, $\Delta G = 24.0 - 200(0.0800) = +8.0\ \mathrm{kJ\,mol^{-1}}$.

Step 2: At 400 K, $\Delta G = 24.0 - 400(0.0800) = -8.0\ \mathrm{kJ\,mol^{-1}}$.

Step 3: The sign changes at $T = \Delta H/\Delta S = 300\ \mathrm{K}$.

Common mistake: (A) reverses the subtraction; (B) adds the entropy term; (D) mixes joules and kilojoules in the wrong direction.

Q19

Chemistry · medium

Aldehydes, ketones, and carboxylic acids / Chemical reactions: Tollens and iodoform tests; mixture analysis

Answer: (B) $1/2$.

Given: Identical portions are tested, so the inferred amounts can be compared directly.

Formula/Principle: One mole of aldehyde reduces two moles of silver ions; one mole of this methyl ketone gives one mole of iodoform.

Step 1: Silver amount is $2.16/108 = 0.0200\ \mathrm{mol}$, so propanal amount is $0.0200/2 = 0.0100\ \mathrm{mol}$.

Step 2: Propanone amount is $3.94/394 = 0.0100\ \mathrm{mol}$.

Step 3: $x_{\mathrm{propanone}} = 0.0100/(0.0100 + 0.0100) = 1/2$.

Common mistake: (A) treats one silver atom as one aldehyde molecule; (C) introduces an extra factor of two in the ketone amount; (D) divides the ketone fraction by two without a stoichiometric basis.

Q20

Chemistry · hard

Electrochemistry / Nernst equation coupled to solubility equilibrium

Answer: (D) $0.74\ \mathrm{V}$.

Given: The copper-ion concentration is controlled by precipitation equilibrium, not fixed at 1 M.

Formula/Principle: $K_{sp} = [\mathrm{Cu^{2+}}][\mathrm{OH^-}]^2$; for $\mathrm{Zn} + \mathrm{Cu^{2+}} \longrightarrow \mathrm{Zn^{2+}} + \mathrm{Cu}$, $n = 2$.

Step 1: $[\mathrm{Cu^{2+}}] = 10^{-20}/(10^{-4})^2 = 10^{-12}\ \mathrm{M}$.

Step 2: $Q = [\mathrm{Zn^{2+}}]/[\mathrm{Cu^{2+}}] = 10^{12}$.

Step 3: $E = 1.10 - (0.060/2)\log(10^{12}) = 1.10 - 0.36 = 0.74\ \mathrm{V}$.

Common mistake: (A) reverses the Nernst sign; (B) assumes standard concentrations; (C) omits the two-electron divisor.

Q21

Chemistry · medium

Organic compounds containing nitrogen / Amines and diazonium salts: nitrous-acid reactions and thermal stability

Answer: (C) $0.010\ \text{mol}$ first and an additional $0.020\ \text{mol}$ on warming.

Why: Primary aliphatic diazonium intermediates are unstable and release nitrogen in the cold reaction. Aniline forms an aromatic diazonium salt that can be retained at 0–5 °C; the stipulated warming then releases its nitrogen. Each amine molecule contributes one molecule of nitrogen gas over the relevant stage.

Why not the others:

(A) Ignores the cold stability of the aromatic diazonium salt.

(B) Reverses which component evolves gas first.

(D) Incorrectly retains the unstable aliphatic intermediate.

Key point: Temperature distinguishes the timing of nitrogen evolution, not the final total number of nitrogen molecules.

Q22

Chemistry · easy

Redox Reactions / Oxidation numbers and charge balance in mixed-valence compounds

Answer: (A) $2/3$.

Why: Four oxide ions contribute charge −8. If x of the three iron ions are Fe(III), the others are Fe(II), so $3x + 2(3 - x) = 8$. Thus $x = 2$, giving two Fe(III) ions out of three iron ions. The fractional average oxidation number does not imply that each individual iron ion has oxidation state 8/3.

Why not the others:

(B) Is the Fe(II) fraction.

(C) Does not supply total charge +8.

(D) Would supply +9.

Key point: An average oxidation number can represent a mixture of integer oxidation states.

Q23

Chemistry · medium

Chemical Kinetics / Integrated first-order rate equation and fractional conversion

Answer: (B) Approximately $38.2\ \text{min}$ additional time.

Given: The first interval leaves 75% of the starting reactant; the target leaves 25%.

Formula/Principle: For first order, $kt = \ln([R]_i/[R]_f)$.

Step 1: $k = \ln(1/0.75)/10 = 0.0288\ \text{min}^{-1}$.

Step 2: The additional interval takes concentration from 0.75 to 0.25 of the original, a ratio of 3.

Step 3: $\Delta t = \ln 3/k = 1.099/0.0288 \approx 38.2\ \text{min}$.

Common mistake: (A) assumes constant-rate percentage loss; (C) is the total time from the beginning; (D) confuses a remaining-fraction ratio with a number of equal time intervals.

Q24

Chemistry · hard

Aldehydes, ketones, and carboxylic acids / Carbonyl reduction and selective functional-group transformations; alcohol acylation

Answer: (C) $C_8H_{14}O_2$, with four possible configurational stereoisomers.

Why: The starting ketone has formula $C_6H_{10}O$ and one stereocentre at the carbon bearing the methyl branch. Ketone reduction adds two hydrogens to give $C_6H_{12}O$ and creates a second stereocentre at the former carbonyl carbon. Alcohol acylation replaces its hydroxyl hydrogen with $COCH_3$, a net addition of C_2H_2O, yielding $C_8H_{14}O_2$. There are two independent stereocentres and no internal symmetry, so all four configurations are distinct. The terminal alkene has two identical hydrogens on one carbon and supplies no E/Z pair.

Why not the others:

(A) Omits the two hydrogens gained in carbonyl reduction.

(B) Counts only one stereocentre, or only products from one starting enantiomer.

(D) Incorrectly doubles the count for E/Z isomerism of a terminal alkene.

Key point: Track elemental changes, newly created stereocentres and the eligibility for geometric isomerism separately.

Q25

Chemistry · medium

Organic Chemistry – Some Basic Principles and Techniques / Isomerism: chirality, racemates and stereochemical consequences of carbonyl formation

Answer: (D) A racemate is present initially and is expected again after achiral reduction.

Why: Butan-2-ol molecules are chiral, but equal amounts of opposite enantiomers cancel in bulk optical rotation. Oxidation makes a trigonal-planar carbonyl carbon and removes that stereocentre. In the stated achiral environment, attack on the two equivalent faces does not preferentially generate one enantiomer.

Why not the others:

(A) Confuses zero net rotation with molecular achirality.

(B) Ignores conversion of tetrahedral carbon to a planar carbonyl.

(C) Invents enantioselectivity without a chiral influence.

Key point: Optical inactivity can be a property of a mixture, not of each molecule.

Q26

Chemistry · medium

Solutions / Osmotic pressure, dissociation and the van't Hoff factor

Answer: (B) Left to right; extra pressure $0.050RT$ atm on the right.

Given: $\alpha = 0.75$ for a salt yielding three ions on full dissociation.

Formula/Principle: $i = 1 + (3-1)\alpha$ and $\Pi = iCRT$.

Step 1: $i = 1 + 2(0.75) = 2.50$. Effective particle concentration on the right is $2.50(0.060) = 0.150\ \text{mol}\,\text{L}^{-1}$.

Step 2: The left has 0.100 M particles, so water moves to the right.

Step 3: $\Delta\Pi = (0.150 - 0.100)RT = 0.050RT$ atm, applied on the right.

Common mistake: (A) compares formula-unit concentrations only; (C) assumes full dissociation; (D) substitutes mass concentration for particle concentration.

Q27

Chemistry · medium

Some Basic Concepts of Chemistry / Mole concept and stoichiometry: decomposition followed by neutralisation

Answer: (A) 30.0 mL.

Given: The final 1.59 g residue is entirely anhydrous sodium carbonate.

Formula/Principle: $2\,\text{NaHCO}_3 \longrightarrow \text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$; $\text{Na}_2\text{CO}_3 + 2\,\text{HCl} \longrightarrow 2\,\text{NaCl} + \text{CO}_2 + \text{H}_2\text{O}$.

Step 1: Residue amount is $1.59/106 = 0.0150$ mol. Its origin in the mixture does not affect the final neutralisation stoichiometry.

Step 2: HCl needed is $2(0.0150) = 0.0300$ mol.

Step 3: $V = n/C = 0.0300\ \text{L} = 30.0$ mL.

Common mistake: (B) stops at one proton per carbonate; (C) doubles the requirement again; (D) incorrectly treats the original 1.90 g as if it were the carbonate residue.

Q28

Chemistry · medium

Classification of Elements and Periodicity in Properties / Electronic configurations and ionic-radius trends in isoelectronic species

Answer: (D) $\text{S}^{2-} > \text{Cl}^{-} > \text{K}^{+} > \text{Ca}^{2+}$ in ionic radius.

Why: Each ion has 18 electrons. With the electron count and occupied-shell pattern the same, the increasing nuclear charge pulls the electron cloud in more strongly. Nuclear charge rises from S to Ca, so radius decreases in that order.

Why not the others:

(A) Reverses the nuclear-charge trend.

(B) Does not follow the common-electron-count comparison.

(C) Reverses both the anion pair and the cation pair.

Key point: For an isoelectronic series, compare nuclear charge before applying neutral-atom size trends.

Q29

Chemistry · medium

Biomolecules / Carbohydrates: non-reducing disaccharides, hydrolysis and reducing sugars

Answer: (C) $4\ \text{mol}$ of silver.

Why: Hydrolysis of one mole of sucrose yields one mole each of glucose and fructose. Although sucrose is non-reducing, the products are reducing sugars under the stated test conditions; fructose can respond through alkaline interconversion. The question explicitly supplies the two-silver-ion stoichiometry for each monosaccharide, so $n(\text{Ag}) = 2(1+1) = 4\ \text{mol}$.

Why not the others:

(A) Carries sucrose's intact non-reducing behaviour over to its hydrolysate.

(B) Counts only one product.

(D) Uses one silver ion for only one of the two monosaccharides.

Key point: Hydrolysis changes accessible reducing functionality.

Q30

Chemistry · hard

Equilibrium / Ionic equilibrium: neutralisation and a weak-base/conjugate-acid buffer

Answer: (A) Approximately 8.52.

Given: Initial ammonia amount $0.100(0.40) = 0.040\ \text{mol}$; first HCl amount $0.100(0.10) = 0.010\ \text{mol}$.

Formula/Principle: Perform strong-acid neutralisation first, then use $pOH = pK_b + \log([\text{NH}_4{}^+]/[\text{NH}_3])$.

Step 1: After the first addition, 0.030 mol ammonia remains and 0.010 mol ammonium is present.

Step 2: The next 0.020 mol HCl leaves 0.010 mol ammonia and produces a total 0.030 mol ammonium; no strong acid remains in excess.

Step 3: Both buffer species share the same final volume, so their concentration ratio is $0.030/0.010 = 3$.

Step 4: $pOH = 5.000 + 0.477 = 5.477$; $pH = 14.000 - 5.477 = 8.523$.

Common mistake: (B) uses the buffer ratio before the second addition; (C) mistakes pKb for pH; (D) assumes partial neutralisation automatically gives a neutral solution.

Mathematics

Method, reasoning and common errors

Q31

Mathematics · easy

Matrices & determinants / Adjoint and inverse of a matrix

Answer: (C) 2.

Given: $A = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}$.

Formula/Principle: $A^{-1} = \operatorname{adj}(A) / \det A$.

Step 1: $\det A = 3(1) - 1(1) = 2 \neq 0$.

Step 2: $A^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 3 \end{pmatrix}$.

Step 3: The diagonal sum is $1/2 + 3/2 = 2$.

Common mistake: (A) uses the original trace; (B) reports only the inverse-determinant factor; (D) introduces an unwarranted negative sign.

Q32

Mathematics · medium

Basic counting techniques / Permutations and Combinations: constrained arrangements

Answer: (B) 96.

Given: Six positions, with B exactly two positions to the right of A.

Formula/Principle: Choose the allowed positions of the constrained pair, then permute all remaining distinct folders.

Step 1: A can occupy position 1, 2, 3 or 4; B then occupies 3, 4, 5 or 6, respectively. There are four ordered position pairs.

Step 2: The other four folders occupy the remaining positions in $4! = 24$ ways.

Step 3: The total is $4 \times 24 = 96$.

Common mistake: (A) loses half the allowed position pairs; (C) incorrectly allows B to the left of A as well; (D) treats A and B as a single adjacent block, which they are not.

Q33

Mathematics · hard

Sets and Logic / Relations & functions: functional equations combined with minimisation

Answer: (D) $-2/3$.

Given: $f(x)+2f(1-x)=x^2$ for all real x.

Formula/Principle: Use the equation at x and at $1-x$ to eliminate the second function value.

Step 1: Substitution gives $f(1-x)+2f(x)=(1-x)^2$.

Step 2: Twice this new equation minus the original gives $3f(x)=2(1-x)^2-x^2=x^2-4x+2$.

Step 3: Hence $f(x)=((x-2)^2-2)/3$.

Step 4: The point x = 2 belongs to the closed interval and makes the square zero, so the minimum is $-2/3$.

Common mistake: (A) assumes a nonnegative right-hand side forces nonnegative f; (B) loses the constant's sign; (C) loses a factor during elimination.

Q34

Mathematics · medium

Complex numbers, linear and quadratic equations / Complex Numbers: geometric loci in the Argand plane

Answer: (A) -1.

Given: Write $z=x+iy$.

Formula/Principle: Equal distances from +1 and −1 define the perpendicular bisector of those points.

Step 1: $(x-1)^2+y^2=(x+1)^2+y^2$ gives x = 0.

Step 2: The other condition becomes $(y-2)^2=5$, with roots $y=2+\sqrt{5}$ and $2-\sqrt{5}$.

Step 3: Their product is $(2+\sqrt{5})(2-\sqrt{5})=4-5=-1$.

Common mistake: (B) takes the magnitude of the signed product; (C) ignores the circle's vertical displacement; (D) multiplies only the shared centre coordinate.

Q35

Mathematics · medium

Integration / Definite-integral symmetry and substitution

Answer: (C) $\pi^2/4$.

Given: The weight $g(x)=\sin x/(1+\cos^2 x)$ satisfies $g(\pi-x)=g(x)$.

Formula/Principle: Reflection of the limits gives $2I=\pi\int_0^{\pi} g(x)\,dx$.

Step 1: Replacing x by $\pi-x$ yields $I=\int_0^{\pi}(\pi-x)g(x)\,dx$; add this to the original integral.

Step 2: With $u=\cos x$, $\int_0^{\pi} g(x)\,dx=\int_{-1}^{1} du/(1+u^2)=\pi/2$.

Step 3: Thus $I=(\pi/2)(\pi/2)=\pi^2/4$.

Common mistake: (A) evaluates only the weight integral; (B) omits the division by two from reflection; (D) divides by two twice.

Q36

Mathematics · medium

Three-dimensional geometry / Cartesian equation of a plane; normal projection and reflection

Answer: (D) $\sqrt{2}$.

Given: A normal to the plane is $\mathbf{n} = (1, 1, 1)$.

Formula/Principle: Reflection in $\mathbf{n} \cdot \mathbf{r} = d$ is $\mathbf{r}' = \mathbf{r} - 2(\mathbf{n} \cdot \mathbf{r} - d)\mathbf{n}/|\mathbf{n}|^2$.

Step 1: Here $\mathbf{n} \cdot P - d = 6 - 3 = 3$ and $|\mathbf{n}|^2 = 3$.

Step 2: Therefore $P' = (1, 2, 3) - 2(1, 1, 1) = (-1, 0, 1)$.

Step 3: $|OP'| = \sqrt{(-1)^2 + 0^2 + 1^2} = \sqrt{2}$.

Common mistake: (A) uses the original point; (B) uses the projection foot (0,1,2), not the reflection; (C) uses twice the perpendicular plane distance, not the origin distance.

Q37

Mathematics · medium

Differentiation / Differentiability of absolute-value functions and one-sided derivatives

Answer: (B) -4.

Given: Only x = 1 and x = −1 can be cusp points.

Formula/Principle: A term $c|x - x_0|$ has a derivative jump of $2c$; differentiability requires the total jump to vanish.

Step 1: Near x = 1, $|x^2 - 1| = |x - 1||x + 1|$ has leading cusp $2|x - 1|$. The other nonsmooth contribution there is $a|x - 1|$, so $a = -2$.

Step 2: Similarly, the cusp at x = −1 has coefficient $2 + b$, so $b = -2$. Terms centred at the other point are smooth locally.

Step 3: $a + b = -4$.

Common mistake: (A) reinforces rather than cancels the cusps; (C) cancels just one cusp; (D) treats cancellation of the coefficients' sum as enough at two distinct locations.

Q38

Mathematics · medium

Statistics, Probability & Linear Programming / Statistics: mean and population variance of combined data

Answer: (A) 9.

Given: Equal group sizes; means 2 and 6; variances 1 and 9.

Formula/Principle: Combined variance equals the average within-group variance plus the variance of group means about the combined mean, with weights proportional to group sizes.

Step 1: Combined mean is $\mu = (2 + 6)/2 = 4$.

Step 2: Within-group contribution is $(1 + 9)/2 = 5$.

Step 3: Between-group contribution is $((2 - 4)^2 + (6 - 4)^2)/2 = 4$. Total variance is $5 + 4 = 9$.

Common mistake: (B) omits between-group variation; (C) adds both squared mean offsets without averaging; (D) fails to normalise the final variance by the combined population size correctly.

Q39

Mathematics · easy

Limit & continuity / Limits of algebraic functions: rationalisation

Answer: (D) 2.

Given: Both square roots are defined in a neighbourhood of zero.

Formula/Principle: Multiply the numerator and denominator by the conjugate.

Step 1: The numerator product becomes $(1+3x)-(1-x)=4x$.

Step 2: For nonzero x near zero, the quotient equals $4/(\sqrt{1+3x}+\sqrt{1-x})$.

Step 3: Letting x tend to zero gives $4/(1+1)=2$.

Common mistake: (A) omits the conjugate denominator; (B) loses a factor of two; (C) substitutes into the indeterminate numerator without treating the simultaneous zero denominator.

Q40

Mathematics · hard

Statistics, Probability & Linear Programming / Probability with permutations and inclusion-exclusion

Answer: (C) $3/16$.

Given: There are $6!=720$ equally likely assignments.

Formula/Principle: Choose the two correct assignments, then require that none of the other four is correct.

Step 1: Choose the two correctly assigned keys in $\binom{6}{2}=15$ ways.

Step 2: By inclusion-exclusion, the number of permutations of the remaining four keys with no fixed assignment is $D_4=4!-\binom{4}{1}3!+\binom{4}{2}2!-\binom{4}{3}1!+\binom{4}{4}0!=24-24+12-4+1=9$.

Step 3: Probability $=15(9)/720=3/16$.

Common mistake: (A) counts only four-cycles, omitting the three ways to form two disjoint swaps; (B) counts only the choice of the correct pair, omitting arrangements of the other keys; (D) allows all 4! assignments of the remainder and hence counts assignments with extra correct keys, some repeatedly.

Q41

Mathematics · medium

Trigonometric functions / Inverse trigonometric functions: addition formula and principal values

Answer: (A) $3\pi/4$.

Given: Each inverse tangent is in $(0,\pi/2)$, and each exceeds $\pi/4$.

Formula/Principle: $\tan(\alpha+\beta)=(\tan\alpha+\tan\beta)/(1-\tan\alpha\tan\beta)$.

Step 1: The tangent of the sum is $(2+3)/(1-6)=-1$.

Step 2: The sum is greater than $\pi/2$ and less than π, so it lies in the second quadrant.

Step 3: The second-quadrant angle with tangent -1 is $3\pi/4$.

Common mistake: (B) ignores the sum's quadrant; (C) loses the sign of the tangent denominator; (D) adds tangent arguments without the denominator correction.

Q42

Mathematics · medium

Differential equations / Solution by separation of variables with an initial condition

Answer: (B) 3.

Given: $y' = y/[x(1+x)]$, with $y(1) = 2$.

Formula/Principle: Separate variables and use $1/[x(1+x)] = 1/x - 1/(1+x)$.

Step 1: $\int dy/y = \int (1/x - 1/(1+x))\,dx$, so $y = Cx/(1+x)$ on the positive solution branch.

Step 2: $2 = C/2$, hence C = 4.

Step 3: $y(3) = 4(3)/4 = 3$.

Common mistake: (A) loses a factor when imposing the initial condition; (C) omits the final denominator change; (D) mistakes the integration constant for the requested function value.

Q43

Mathematics · medium

Complex numbers, linear and quadratic equations / Quadratic Equations: location of real roots

Answer: (C) $a > 3$.

Why: The monic quadratic opens upwards. A test point lies strictly between its two real roots exactly when the polynomial is negative there. At x = 1, the value is $1 - 2a + a + 2 = 3 - a$, so the required inequality is $3 - a < 0$. This also guarantees two distinct real roots because a continuous upward-opening quadratic is below zero at 1 and above zero for sufficiently large positive and negative x.

Why not the others:

(A) Includes a = 3, when 1 is a root rather than strictly between the roots.

(B) Only enforces distinct-real-root discriminant conditions.

(D) Reverses the sign test.

Key point: Root location requires more than a nonnegative discriminant.

Q44

Mathematics · hard

Integration / Parameter-dependent absolute-value integral and minimisation

Answer: (D) $1/4$.

Given: For a in [0,1], the sign changes at $x = \sqrt{a}$.

Formula/Principle: Split the integral at its changing zero before differentiating the resulting expression.

Step 1: $F(a) = \int_0^{\sqrt{a}} (a - x^2)\,dx + \int_{\sqrt{a}}^1 (x^2 - a)\,dx$.

Step 2: Evaluation gives $F(a) = 1/3 - a + (4/3)a^{3/2}$.

Step 3: $F'(a) = -1 + 2\sqrt{a}$, which is negative below a = 1/4 and positive above it. Thus a = 1/4 is the minimum point.

Step 4: $F(1/4) = 1/3 - 1/4 + (4/3)(1/8) = 1/4$. The endpoint values are 1/3 and 2/3, both larger.

Common mistake: (A) uses the endpoint a = 0; (B) drops one of the two positive area pieces; (C) treats the integrand's sign change as a fixed midpoint without optimising the parameter.

Q45

Mathematics · medium

Vectors / Dot/cross products and area of a parallelogram

Answer: (B) $3\sqrt{3}/2$ square units.

Given: $|\mathbf{a}| = |\mathbf{b}| = 1$ and the included angle is 60°.

Formula/Principle: Parallelogram area is the magnitude of the cross product of its adjacent sides.

Step 1: Expand $(\mathbf{a} + 2\mathbf{b}) \times (\mathbf{a} - \mathbf{b}) = -\mathbf{a} \times \mathbf{b} + 2\mathbf{b} \times \mathbf{a} = -3\mathbf{a} \times \mathbf{b}$.

Step 2: $|\mathbf{a} \times \mathbf{b}| = 1(1)\sin 60^\circ = \sqrt{3}/2$.

Step 3: The requested area is $3\sqrt{3}/2$.

Common mistake: (A) omits the linear-combination factor; (C) uses sin 60° = 1; (D) calculates triangle area instead of parallelogram area.

Physics

Method, reasoning and common errors

Q46

Physics · medium

Motion in a Straight Line / Distance and displacement from a signed velocity-time graph

Answer: (A) Average speed 2 m/s; average velocity zero.

Given: The piecewise linear graph crosses v = 0 at t = 3 s.

Formula/Principle: Displacement is signed area under v(t); distance is the area under |v(t)|.

Step 1: From 0 to 2 s, the triangular area is 4 m. From 2 to 3 s, it is another 2 m.

Step 2: The negative-velocity intervals 3 to 4 s and 4 to 6 s contribute signed areas -2 m and -4 m.

Step 3: Displacement is zero; distance is 4 + 2 + 2 + 4 = 12 m. Divide each by 6 s to get 0 and 2 m/s.

Common mistake: (B) interchanges speed and velocity; (C) misses area beside the central zero crossing; (D) treats negative velocity as positive when calculating displacement.

Q47

Physics · easy

Units and Measurements / Dimensional consistency of an empirical pressure model

Answer: (C) $ML^{-2}T^{-2}$.

Why: Pressure has dimensions force/area, or $ML^{-1}T^{-2}$. Every additive term must have pressure dimensions. Since h has dimensions L, $[b] = [P]/[h] = ML^{-2}T^{-2}$. The first term is also consistent because $[\rho v^2] = (ML^{-3})(L^2T^{-2})$.

Why not the others:

(A) Gives pressure dimensions without dividing by height.

(B) Gives acceleration dimensions, omitting density.

(D) Gives density dimensions, omitting acceleration.

Key point: Dimensional consistency applies separately to each additive term.

Q48

Physics · medium

Mechanical Properties of Fluids / Continuity and Bernoulli equation in a horizontal pipe

Answer: (B) $(15/2)\rho v^2$.

Given: The radius decreases by a factor of two, while elevation remains constant.

Formula/Principle: $A_1 v_1 = A_2 v_2$ and $P + \rho v^2/2$ is constant along a streamline.

Step 1: Area decreases by a factor of four, so the narrow-section speed is 4v.

Step 2: $P_1 - P_2 = \rho((4v)^2 - v^2)/2 = (15/2)\rho v^2$.

Common mistake: (A) doubles speed instead of quadrupling it; (C) omits the upstream kinetic-energy term; (D) mixes the area ratio with the kinetic-energy expression.

Q49

Physics · hard

System of Particles and Rotational Motion / Angular momentum conservation and actuator work

Answer: (D) $4\omega/3$ and $mR^2\omega^2/2$.

Given: No external axial torque, but the actuator may supply mechanical energy.

Formula/Principle: $I_i\omega = I_f\omega_f$; supplied work equals the change in total mechanical kinetic energy.

Step 1: $I_i = 3mR^2$ and $I_f = 2mR^2 + m(R/2)^2 = 9mR^2/4$.

Step 2: $\omega_f = (3/(9/4))\omega = 4\omega/3$.

Step 3: $K_i = 3mR^2\omega^2/2$, while $K_f = \frac{1}{2}(9mR^2/4)(4\omega/3)^2 = 2mR^2\omega^2$.

Step 4: Supplied work $K_f - K_i = mR^2\omega^2/2 > 0$.

Common mistake: (A) reverses the inertia ratio; (B) confuses conservation of angular momentum with conservation of kinetic energy; (C) assumes fixed angular speed without an external torque.

Q50

Physics · medium

Thermodynamics / Heat-engine energy balance and the Carnot bound

Answer: (A).

Why: The rejected heat is 800 - 300 = 500 J, satisfying the cyclic first-law balance. Actual efficiency is $300/800 = 0.375$. The maximum reversible efficiency is $1 - 300/500 = 0.40$. The given efficiency is below the bound, consistent with irreversibility. The reservoir entropy change is $500/300 - 800/500 = 1/15\,\mathrm{J\,K^{-1}} > 0$.

Why not the others:

(B) A reversible engine would attain 0.40 efficiency, not 0.375.

(C) Energy is conserved by the stated heat rejection.

(D) 0.375 is less than 0.40.

Key point: Test energy balance and the second-law limit separately.

Q51

Physics · medium

Oscillations / Spring force in vertical simple harmonic motion

Answer: (B) $3mg/2$.

Given: Equilibrium extension is 0.10 m; oscillation amplitude about equilibrium is 0.05 m.

Formula/Principle: At equilibrium $k(0.10) = mg$; spring tension is k times extension from the natural length.

Step 1: Maximum extension is $0.10 + 0.05 = 0.15$ m.

Step 2: Maximum tension $= k(0.15) = 1.5mg$. The minimum extension is positive, so the spring remains in tension.

Common mistake: (A) uses the amplitude alone, giving the oscillating net-force amplitude; (C) doubles the equilibrium tension without using the actual amplitude; (D) assumes tension always equals weight.

Q52

Physics · medium

Waves / Resolving the ambiguity in a beat-frequency measurement

Answer: (C) 508 Hz.

Given: The initial frequency difference is 4 Hz; after the reference frequency decreases by 2 Hz, the difference is 2 Hz.

Formula/Principle: Beat frequency equals the absolute difference of the two frequencies.

Step 1: Initially the unknown could be $512 - 4 = 508$ Hz or $512 + 4 = 516$ Hz.

Step 2: Relative to 510 Hz, these possibilities give 2 Hz and 6 Hz beats, respectively.

Step 3: Only 508 Hz agrees with both observations.

Common mistake: (A) subtracts the new difference from an already shifted unknown estimate; (B) reports the modified reference frequency; (D) uses only the first observation.

Q53

Physics · medium

Electric Charges and Fields / Work from a nonuniform electric-field graph and a turning point

Answer: (D) $7a/4$.

Given: The electric force opposes the initial motion throughout the shown positive-x region.

Formula/Principle: $K(x) - K(0) = q\int_0^x E_x(s)\,ds$.

Step 1: From 0 to a, the signed triangular field area is $-E_0a/2$, so $K(a) = qE_0a(5/4 - 1/2) = 3qE_0a/4$.

Step 2: Beyond a, the field is constant at $-E_0$. Stopping requires $qE_0(x - a) = 3qE_0a/4$.

Step 3: Thus $x = a + 3a/4 = 7a/4$, inside the indicated interval.

Common mistake: (A) uses the full initial energy with a constant field from x = 0; (B) miscounts the triangular area; (C) assumes the end of the drawn graph is a physical stopping boundary.

Q54

Physics · hard

Current Electricity / An unbalanced resistor bridge with an ideal ammeter

Answer: (B) $12/17$ A from A to B.

Given: The ideal ammeter makes A and B equipotential, but it can carry a nonzero current.

Formula/Principle: Apply Ohm's law and Kirchhoff's current law after accounting for the ammeter connection.

Step 1: Above the common middle node, $R_u = 2 \parallel 3 = 6/5\,\Omega$. Below it, $R_l = 4 \parallel 3 = 12/7\,\Omega$.

Step 2: The supply current is $12/(6/5 + 12/7) = 70/17$ A. The middle-node potential relative to the bottom is $(70/17)(12/7) = 120/17$ V.

Step 3: Current arriving at A through 2 ohm is $(12 - 120/17)/2 = 42/17$ A; current leaving A through 4 ohm is $30/17$ A.

Step 4: The excess $12/17$ A flows from A through the meter to B.

Common mistake: (A) uses the disconnected voltage-divider difference as a current; (C) reverses Kirchhoff's excess-current direction; (D) confuses zero meter voltage with zero meter current.

Q55

Physics · easy

Nuclei / Mass number and atomic number in alpha and beta-minus decay

Answer: (A) 234, 92.

Why: Alpha emission reduces mass number by four and atomic number by two, giving (234,90). Each beta-minus decay converts a neutron into a proton, increasing atomic number by one while preserving mass number. After two such decays the pair is (234,92).

Why not the others:

(B) Stops after the alpha decay.

(C) Accounts for beta-minus changes but omits alpha emission.

(D) Incorrectly decreases mass number by two rather than four in alpha decay.

Key point: Beta-minus emission changes the proton-neutron composition without changing the nucleon count.

Q56

Physics · medium

Electromagnetic Induction / Motional emf and terminal speed on inclined rails

Answer: (D) $mgR\sin\theta/(B^2l^2)$.

Given: Field is normal to the plane of motion and the total resistance is constant.

Formula/Principle: Motional emf $\mathcal{E} = Blv$; current $I = Blv/R$; magnetic braking force $F_B = BIl$.

Step 1: $F_B = B^2l^2v/R$, directed up the incline by Lenz's law.

Step 2: At terminal speed the net force along the rails is zero: $mg\sin\theta = B^2l^2v_t/R$.

Step 3: Solve to obtain the stated expression.

Common mistake: (A) uses the normal component of gravity; (B) inserts an inappropriate kinetic-energy factor of two into a force balance; (C) doubles the motional emf or gravitational component without justification.

Q57

Physics · medium

Ray Optics and Optical Instruments / Apparent depth through two plane liquid layers

Answer: (C) 7 cm; unchanged.

Given: Layer depths are 6 cm and 4 cm, with indices 3/2 and 4/3; the viewing medium is air.

Formula/Principle: In the paraxial approximation, apparent depth through parallel layers seen from air is $d_{\text{app}} = \sum_i t_i/n_i$.

Step 1: The contributions are $6/(3/2) = 4$ cm and $4/(4/3) = 3$ cm.

Step 2: Total apparent depth is 7 cm.

Step 3: Reversing layer order leaves the same two terms in the sum, so it does not change the result under the stated conditions.

Common mistake: (A) ignores refraction; (B) incorrectly treats interface order as changing the paraxial sum for fixed layer depths; (D) multiplies depths by indices instead of dividing.

Q58

Physics · medium

Alternating Current / Series RLC phasors and real power

Answer: (A) 10 V and 16 W.

Given: The reactive voltages are opposite in phase and the resistor voltage is in phase with current.

Formula/Principle: $V = \sqrt{V_R^2 + (V_L - V_C)^2}$; average power $P = IV_R$ for ideal L and C.

Step 1: Net reactive voltage is 10 - 4 = 6 V.

Step 2: Source voltage $= \sqrt{8^2 + 6^2} = 10$ V.

Step 3: Average power $= 2(8) = 16$ W; equivalently R = 4 ohm and $I^2R = 16$ W.

Common mistake: (B) adds all voltage magnitudes and equates apparent with real power; (C) uses $VI = 20$ VA as watts without a power factor; (D) combines the resistor and inductor voltages as in-phase scalars.

Q59

Physics · medium

Semiconductor Electronics: Materials, Devices and Simple Circuits / Half-wave rectification and mean power in a resistor

Answer: (B) $1/2$.

Given: The diode transmits one half-cycle unchanged and blocks the other.

Formula/Principle: Average resistor power is $\langle v^2\rangle/R$, not $\langle v\rangle^2/R$.

Step 1: For the direct sine wave, $\langle v^2\rangle = V_0^2/2$, so $P_{\text{direct}} = V_0^2/(2R)$.

Step 2: Each half-cycle contributes equal squared-voltage area. Removing one gives $P_{\text{half}} = V_0^2/(4R)$.

Step 3: The power ratio is 1/2.

Common mistake: (A) reports an rms-voltage ratio rather than a power ratio; (C) halves an already averaged voltage and squares it; (D) ignores that half the cycle carries no current.

Q60

Physics · hard

Wave Optics / Two-colour interference, phase reversal and minimum slab thickness

Answer: (D) $3.0\,\mu\text{m}$.

Given: Each colour begins with a relative phase of π; slab excess optical path is $\delta = (n-1)t = t/2$.

Formula/Principle: A bright fringe requires $\pi + 2\pi\delta/\lambda = 2\pi k$, so δ must be an odd multiple of $\lambda/2$ for each colour.

Step 1: We need $\delta = (2r+1)300\,\text{nm} = (2s+1)500\,\text{nm}$, with nonnegative integers r and s.

Step 2: The smallest common value with odd multipliers is $\delta = 1500$ nm: five times 300 and three times 500.

Step 3: $t = 1500/0.50 = 3000$ nm, or 3.0 micrometres.

Common mistake: (A) satisfies only the 600 nm condition; (B) uses an integer-wavelength condition appropriate to no initial phase reversal and still misses the second colour; (C) again satisfies only the 600 nm odd-half-wavelength condition. The two different colours need not interfere with one another.
